

Federated Nonlinear Causal Discovery via Divide-and-Conquer Learning

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Abstract

Federated causal discovery aims to learn causal structures from distributed data without sharing raw samples. Existing federated nonlinear methods adopt a monolithic global strategy that optimizes the entire graph simultaneously, suffering from *catastrophic error propagation*: a single misidentified edge cascades through the global structure, severely degrading accuracy under heterogeneous and limited local data. We propose DC-FNCD (Divide-and-Conquer based Federated Nonlinear Causal Discovery), which decomposes the global problem into independent per-variable neighborhood learning tasks. The core mechanism is a characteristic function-based conditional independence test whose empirical statistics admit *exact linear decomposition* across clients, enabling *lossless* federated aggregation without raw data exchange. Local neighborhoods are merged via conflict-aware skeleton construction and oriented through federated additive noise model testing. Extensive experiments demonstrate that DC-FNCD significantly outperforms state-of-the-art federated baselines. The source code is available at <https://github.com/Xianjie-Guo/DC-FNCD>.

CCS Concepts

• Computing methodologies → Probabilistic reasoning.

Keywords

Causal Discovery; Federated Learning; Privacy-Preserving Machine Learning; Nonlinear Causal Relationship

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1 Introduction

Background. Causal discovery, the task of uncovering cause-effect relationships from observational data [12, 21, 38], is fundamental to scientific reasoning across medicine [6], genomics [32], and climate science [4]. In practice, the data required for causal discovery is often distributed across autonomous parties (e.g., healthcare networks, financial institutions) and cannot be centralized due to privacy regulations such as the GDPR. Federated learning (FL) [5, 17, 18, 25, 26] addresses this by coordinating model training across clients without accessing raw data, leading to *federated causal discovery* (FCD) [9, 16, 22, 40]. Early FCD methods predominantly assumed linear causal mechanisms, in which structural equation models (SEMs) are parameterized by weighted adjacency matrices and the problem reduces to learning a directed acyclic graph (DAG) via continuous optimization [22] or to conditional independence testing with closed-form statistics [16]. However, real-world causal mechanisms are rarely linear. Consider a multi-hospital study investigating the causal drivers of sepsis mortality: the relationship between inflammatory biomarker levels and organ failure is highly nonlinear and exhibits threshold effects that vary across patient demographics [29]. In such settings, linear models are fundamentally misspecified, and nonlinear FCD becomes essential. The challenge is further compounded when data across clients exhibit heterogeneous distributions, as different hospitals may serve distinct patient populations with varying disease prevalences and comorbidity patterns.

Related Work. Federated nonlinear causal discovery has attracted increasing attention in recent years. FedDAG [7] pioneered gradient-based federated DAG learning under additive noise models (ANMs), employing a two-level architecture that separates mechanism learning (local) from structure learning (global) via continuous acyclicity optimization. NOTEARS-ADMM [22] adapted the NOTEARS framework [41] to the federated setting using the alternating direction method of multipliers (ADMM), supporting both linear and nonlinear (MLP-parameterized) cases by exchanging only model parameters. FedCausal [36] proposed a unified global optimization formula with adaptive handling of heterogeneous data, extending to nonlinear and nonparametric versions. FedCDH [19] developed a constraint-based approach that utilizes federated conditional independence testing via random feature approximation of kernel matrices and a federated independent change principle for orientation, making no assumptions about the functional form. FedPC [16] introduced a layer-wise aggregation strategy that adapts the PC algorithm [30] to federated settings, with a kernel-based variant (FedPC-KBT) for handling nonlinear dependencies. A more comprehensive review covering both linear and nonlinear federated causal discovery methods is provided in [Appendix A](#).

Motivation. The aforementioned methods have made significant strides in enabling causal discovery from distributed data under nonlinear settings. Nevertheless, we identify a fundamental architectural limitation shared by virtually all existing approaches: they adopt a *monolithic global learning strategy* that attempts to reconstruct the entire causal graph simultaneously. Score-based methods (e.g., FedDAG) optimize a single global objective over the full adjacency matrix, where each gradient update entangles all p^2 potential edges through the acyclicity constraint $\text{tr}(e^{W \circ W}) - p = 0$ [41]. Constraint-based methods (e.g., FedCDH) execute a global PC-style algorithm where conditioning sets grow across the entire variable space, with each independence decision depending on the accumulated history of all prior decisions. This global coupling introduces *catastrophic error propagation*: a single misidentified nonlinear relationship corrupts the structural context for all downstream decisions. Consider a DAG $X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4$ learned from heterogeneous FL data. If the nonlinear edge $X_1 \rightarrow X_2$ is erroneously removed in the global skeleton phase due to insufficient local samples at certain clients, then the subsequent conditioning on X_2 during the test $X_1 \perp\!\!\!\perp X_3 \mid X_2$ becomes invalid: the algorithm may now incorrectly infer a direct edge $X_1 \rightarrow X_3$, which in turn distorts the orientation of $X_3 \rightarrow X_4$. This single error cascades through the entire structure. In FL settings where clients hold limited, heterogeneous, and high-dimensional data, such errors are not exceptions but *the norm*, and the monolithic strategy amplifies them into global structural failures (see Figure 1).

Contributions. To address these limitations, we propose DC-FNCD (Divide-and-Conquer based Federated Nonlinear Causal Discovery), a fundamentally new architecture for federated nonlinear causal discovery that replaces monolithic global learning with *local-to-global structure synthesis*. Our main contributions are:

- (1) **A divide-and-conquer framework for federated nonlinear causal discovery.** We decompose the global structure learning problem into independent local neighborhood learning subproblems, one per variable. Each variable independently discovers

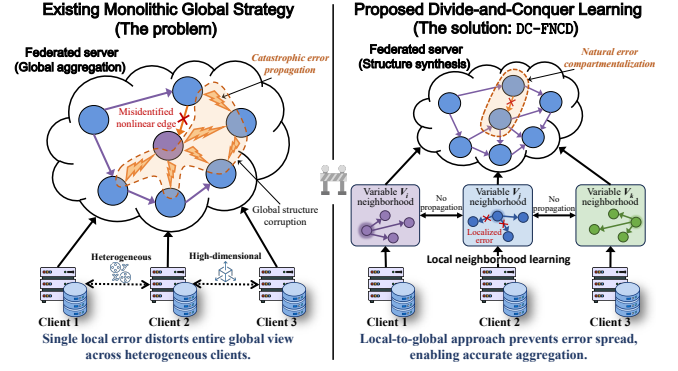


Figure 1: The necessity of a divide-and-conquer strategy for robust federated nonlinear causal discovery.

its parent-child (PC) set via federated conditional independence testing, and the local results are merged through a conflict-aware skeleton construction procedure and oriented via federated ANM-based verification.

- (2) **A novel characteristic function-based federated independence test with exact linear decomposability.** We develop a nonparametric conditional independence test based on the *characteristic function discrepancy*, whose empirical estimator admits *exact* decomposition as a sample-weighted sum of local client statistics. Our characteristic function formulation enables *loss-less* federated aggregation while naturally capturing arbitrary nonlinear dependencies. We establish the asymptotic distribution of the test statistic (Theorem 3.3) and prove its consistency under the federated setting (Lemma 3.4).
- (3) **Rigorous theoretical analysis.** We prove that DC-FNCD recovers the true causal skeleton with high probability under standard causality-based assumptions (Theorem 3.7), provide formal privacy guarantees through characteristic function aggregation (Theorem 3.10), and analyze the communication complexity showing that our method achieves an $O(p)$ -factor reduction compared to global methods (Proposition 3.11).
- (4) **Extensive empirical validation.** We conduct experiments on homogeneous and heterogeneous nonlinear synthetic benchmarks as well as real-world datasets, demonstrating that DC-FNCD significantly outperforms state-of-the-art baselines.

2 Preliminaries

We introduce the notation, definitions, and assumptions used throughout this paper. A comprehensive summary of key notations and their descriptions is provided in Table 1.

Notation. Let $\mathcal{V} = \{V_1, V_2, \dots, V_p\}$ denote a set of p random variables. We consider a federated setting with K clients, where client $k \in [K] \triangleq \{1, 2, \dots, K\}$ holds a private local dataset $\mathbf{D}^{(k)} = \{\mathbf{x}_j^{(k)}\}_{j=1}^{n_k} \subset \mathbb{R}^p$ with n_k samples drawn from a distribution $\mathbb{P}^{(k)}$. The total number of samples across all clients is $N = \sum_{k=1}^K n_k$, and the weight of client k is $w_k = n_k/N$. All communication occurs through summary statistics transmitted to a central server.

Definition 2.1 (Causal DAG and Structural Equation Model). A causal directed acyclic graph (DAG) is a pair $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where

Table 1: Summary of key notations and their descriptions.

Notation	Description
$\mathcal{V} = \{V_1, \dots, V_p\}$	Set of p observable variables
$\mathcal{G}^* = (\mathcal{V}, \mathcal{E}^*)$	True causal DAG
K	Number of federated clients
$\mathbf{D}^{(k)} \in \mathbb{R}^{n_k \times p}$	Local dataset at client k
n_k	Number of samples at client k
$N = \sum_k n_k$	Total sample size across all clients
$w_k = n_k/N$	Weight of client k
$\text{Pa}_{\mathcal{G}}(V_i), \text{Ch}_{\mathcal{G}}(V_i)$	Parent and child sets of V_i in $\mathcal{G}$
$\text{PC}_{\mathcal{G}}(V_i)$	Parent-child (adjacency) set of V_i
$\varphi_X(t)$	Characteristic function of X
$\varphi_{XY}(s, t)$	Joint characteristic function of (X, Y)
$\hat{\phi}_X^{(k)}(t)$	Empirical CF of X at client k
$\Delta_{XY}(s, t)$	CF discrepancy: $\varphi_{XY}(s, t) - \varphi_X(s)\varphi_Y(t)$
$\hat{T}_N$	Federated CF independence test statistic
L	Number of random frequency points
$(s_\ell, t_\ell)_{\ell=1}^L$	Sampled frequency points
σ_ω	Frequency bandwidth parameter
α	Significance level for CI tests
$d_{\max}$	Maximum conditioning set size
B	Number of permutations in permutation test
$R_X = X - \hat{g}_X(Z)$	Nonparametric residual of X given Z
$\hat{f}_{ji}^{(k)}$	Local KNN regression of V_j on V_i at client k
$\mathcal{E}_{\text{sym}}, \mathcal{E}_{\text{asym}}$	Symmetric and asymmetric edge sets
$\text{SepSet}(V_i, V_j)$	Separation set for pair (V_i, V_j)
C_i	Candidate PC set for variable V_i
τ	Confirmation threshold for asymmetric edges
$\hat{T}_{i,j}^{\rightarrow}, \hat{T}_{i,j}^{\leftarrow}$	ANM test statistics for forward/backward directions

$\mathcal{V}$ is a set of nodes (variables) and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of directed edges such that $\mathcal{G}$ contains no directed cycles. The structural equation model (SEM) associated with $\mathcal{G}$ specifies, for each $V_i \in \mathcal{V}$:

$$V_i = f_i(\text{Pa}_{\mathcal{G}}(V_i), \varepsilon_i), \quad \varepsilon_i \sim \mathbb{P}_{\varepsilon_i}, \quad (1)$$

where $\text{Pa}_{\mathcal{G}}(V_i) \subseteq \mathcal{V}$ denotes the parent set of V_i in $\mathcal{G}$, f_i is the causal mechanism, and ε_i is an exogenous noise variable.

Definition 2.2 (Characteristic Function). For a real-valued random variable X with distribution $\mathbb{P}_X$, the characteristic function (CF) is the Fourier transform of $\mathbb{P}_X$:

$$\varphi_X(t) \triangleq \mathbb{E}[e^{itX}] = \mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)], \quad t \in \mathbb{R}. \quad (2)$$

For a random vector $(X, Y) \in \mathbb{R}^2$, the joint characteristic function is:

$$\varphi_{XY}(s, t) \triangleq \mathbb{E}[e^{i(sX+tY)}], \quad (s, t) \in \mathbb{R}^2. \quad (3)$$

A fundamental property of characteristic functions is that they uniquely determine distributions [31], and moreover, they provide a necessary and sufficient condition for statistical independence:

$$X \perp\!\!\!\perp Y \iff \varphi_{XY}(s, t) = \varphi_X(s) \cdot \varphi_Y(t), \quad \forall (s, t) \in \mathbb{R}^2. \quad (4)$$

This equivalence, which follows from the injectivity of the Fourier transform and the factorization of product measures [2], forms the theoretical foundation of our federated independence test.

Definition 2.3 (Conditional Independence via Residualization). Let X, Y, Z be random variables (or vectors). We say X is conditionally independent of Y given Z , denoted $X \perp\!\!\!\perp Y \mid Z$, if $\mathbb{P}(X, Y \mid Z) =$

$\mathbb{P}(X \mid Z) \cdot \mathbb{P}(Y \mid Z)$. Under the nonlinear residualization approach, we define:

$$R_X \triangleq X - g_X(Z), \quad R_Y \triangleq Y - g_Y(Z), \quad (5)$$

where $g_X(Z) = \mathbb{E}[X \mid Z]$ and $g_Y(Z) = \mathbb{E}[Y \mid Z]$ are the conditional expectations. Under mild regularity conditions (see Assumption 5 below), conditional independence can be tested through the marginal independence of the residuals: $X \perp\!\!\!\perp Y \mid Z \implies R_X \perp\!\!\!\perp R_Y$.

We now state the assumptions underpinning our method.

Assumption 1 (Causal Sufficiency [30]). The variable set $\mathcal{V}$ is causally sufficient, meaning there are no latent common causes of any pair of variables in $\mathcal{V}$.

Assumption 2 (Faithfulness [30]). The joint distribution $\mathbb{P}$ over $\mathcal{V}$ is faithful to the true DAG $\mathcal{G}^*$, i.e., every conditional independence in $\mathbb{P}$ corresponds to a d -separation in $\mathcal{G}^*$, and vice versa.

Assumption 3 (Additive Noise Model). The SEM in Eq. (1) takes the additive noise form:

$$V_i = f_i(\text{Pa}_{\mathcal{G}}(V_i)) + \varepsilon_i, \quad \varepsilon_i \perp\!\!\!\perp \text{Pa}_{\mathcal{G}}(V_i), \quad (6)$$

where f_i is a (possibly nonlinear) measurable function and ε_i is independent noise. This model is identifiable under mild conditions on f_i and $\mathbb{P}_{\varepsilon_i}$ [15, 23].

Assumption 4 (Shared Causal Structure [7, 34, 37]). All K clients share the same causal DAG $\mathcal{G}^* = (\mathcal{V}, \mathcal{E}^*)$. The local distributions $\mathbb{P}^{(k)}$ may differ in causal mechanisms $f_i^{(k)}$ and noise distributions $\mathbb{P}_{\varepsilon_i}^{(k)}$, but the qualitative causal structure (i.e., the causal edge set $\mathcal{E}^*$) is invariant across clients.

Assumption 4 is a standard assumption in the field of FCD.

Assumption 5 (Residual Sufficiency for CI Testing). For variables X, Y and conditioning set Z , if $X \perp\!\!\!\perp Y \mid Z$, then the residuals $R_X = X - g_X(Z)$ and $R_Y = Y - g_Y(Z)$ satisfy $R_X \perp\!\!\!\perp R_Y$. This holds when the regression functions g_X, g_Y are correctly specified, i.e., $g_X(Z) = \mathbb{E}[X \mid Z]$ and $g_Y(Z) = \mathbb{E}[Y \mid Z]$, which is guaranteed for consistent nonparametric estimators such as k -nearest neighbors regression in the large-sample limit [14].

3 The Proposed DC-FNCD Method

We present DC-FNCD, a divide-and-conquer framework for federated nonlinear causal discovery. As illustrated in Figure 2, DC-FNCD consists of three synergistic phases: (i) *Federated Local PC Learning* (Section 3.1); (ii) *Conflict-Aware Skeleton Construction* (Section 3.2); and (iii) *Federated ANM-Based Orientation* (Section 3.3).

3.1 Phase 1: Federated Local PC Learning

The core innovation of DC-FNCD lies in decomposing the global causal discovery problem into p independent local subproblems. For each target variable V_i , we learn its parent-child set $\text{PC}_{\mathcal{G}^*}(V_i)$ via an incremental conditional independence testing procedure, executed entirely through federated characteristic function aggregation.

3.1.1 Characteristic Function Discrepancy Test. We introduce a non-parametric independence test based on the *characteristic function discrepancy* (CFD). The fundamental idea exploits the equivalence in Eq. (4): X and Y are independent if and only if their joint characteristic function factorizes as the product of marginals.

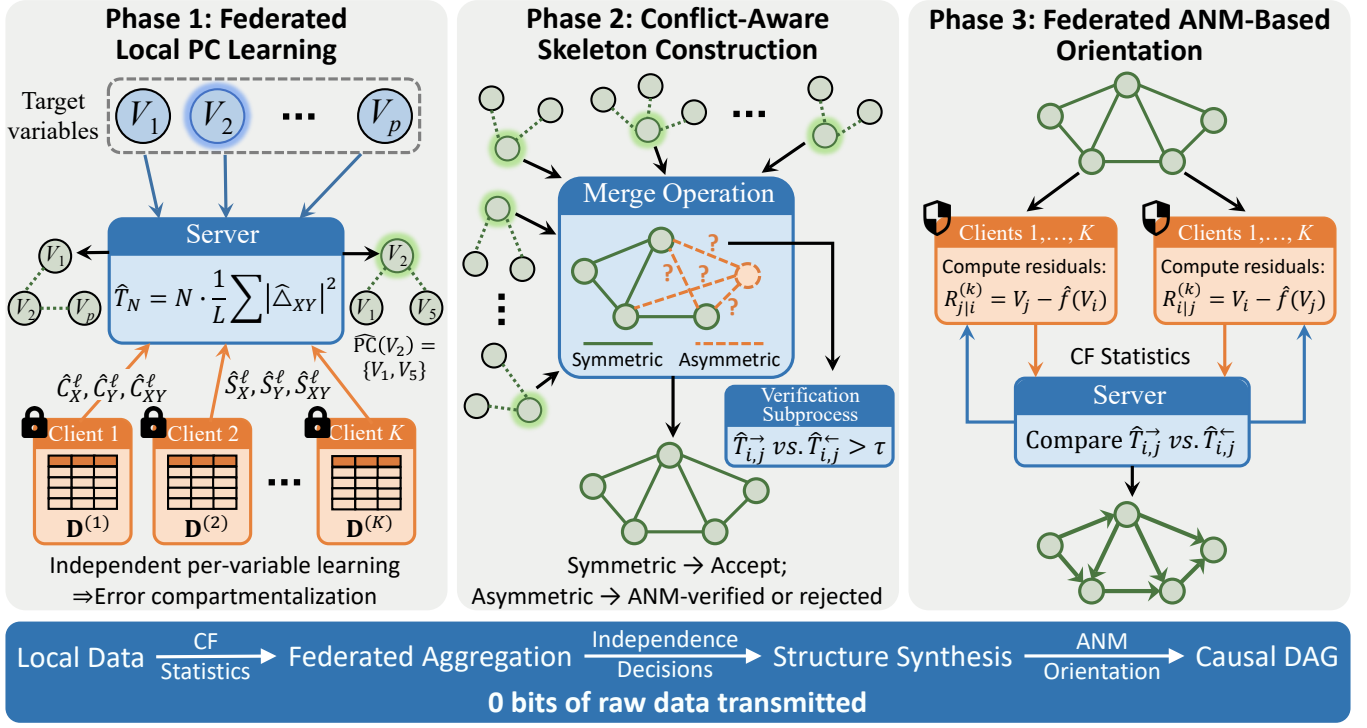


Figure 2: Overview of the DC-FNCD framework. Specifically, Phase 1: Each variable independently learns its PC set through federated characteristic function-based CI testing. Clients compute local CF statistics on private data and transmit only aggregable summaries. Phase 2: The server merges local PC sets into a global skeleton via symmetric confirmation, with asymmetric conflicts resolved through federated ANM-based verification. Phase 3: Remaining undirected edges are oriented using federated ANM testing, comparing residual independence scores in both causal directions.

Definition 3.1 (Characteristic Function Discrepancy). For random variables X and Y , the characteristic function discrepancy at frequency point $(s, t) \in \mathbb{R}^2$ is:

$$\Delta_{XY}(s, t) \triangleq \varphi_{XY}(s, t) - \varphi_X(s) \cdot \varphi_Y(t). \quad (7)$$

The integrated squared discrepancy with respect to a frequency distribution μ on $\mathbb{R}^2$ is:

$$\mathcal{T}(X, Y) \triangleq \int_{\mathbb{R}^2} |\Delta_{XY}(s, t)|^2 d\mu(s, t). \quad (8)$$

We have $\mathcal{T}(X, Y) = 0$ if and only if $X \perp\!\!\!\perp Y$ (by the uniqueness theorem for characteristic functions). In practice, we approximate the integral by Monte Carlo sampling: let $\{(s_\ell, t_\ell)\}_{\ell=1}^L$ be L i.i.d. frequency points drawn from $\mu = \mathcal{N}(0, \sigma_\omega^2) \otimes \mathcal{N}(0, \sigma_\omega^2)$, where $\sigma_\omega > 0$ is a bandwidth parameter. The empirical approximation is:

$$\hat{\mathcal{T}}_L(X, Y) = \frac{1}{L} \sum_{\ell=1}^L |\hat{\Delta}_{XY}(s_\ell, t_\ell)|^2, \quad (9)$$

where $\hat{\Delta}_{XY}(s_\ell, t_\ell)$ is the empirical estimator of the discrepancy at frequency (s_ℓ, t_ℓ) .

Empirical Estimation and Linear Decomposability. Given n i.i.d. samples $\{(x_j, y_j)\}_{j=1}^n$, the empirical characteristic functions are:

$$\hat{\varphi}_X(s) = \frac{1}{n} \sum_{j=1}^n e^{isx_j} = \frac{1}{n} \sum_{j=1}^n [\cos(sx_j) + i \sin(sx_j)], \quad (10)$$

$$\hat{\varphi}_Y(t) = \frac{1}{n} \sum_{j=1}^n e^{ity_j} = \frac{1}{n} \sum_{j=1}^n [\cos(ty_j) + i \sin(ty_j)], \quad (11)$$

$$\hat{\varphi}_{XY}(s, t) = \frac{1}{n} \sum_{j=1}^n [\cos(sx_j + ty_j) + i \sin(sx_j + ty_j)]. \quad (12)$$

A crucial observation is that each empirical characteristic function is a *sample average*, which admits exact linear decomposition across clients. Specifically, for client k holding n_k samples $\{x_j^{(k)}\}_{j=1}^{n_k}$, define the *local characteristic function statistics*:

$$\hat{\varphi}_X^{(k)}(s) \triangleq \frac{1}{n_k} \sum_{j=1}^{n_k} e^{isx_j^{(k)}}. \quad (13)$$

Then the *global* empirical characteristic function is recovered exactly via weighted aggregation:

$$\hat{\varphi}_X(s) = \sum_{k=1}^K w_k \hat{\varphi}_X^{(k)}(s), \quad w_k = \frac{n_k}{N}. \quad (14)$$

This decomposition is *lossless*, i.e., no approximation is introduced by the federated aggregation, and constitutes a key advantage over kernel-based methods that require random feature approximation [27].

We formalize this property in the following lemma.

Lemma 3.2. *Let $\hat{\phi}_X^{(k)}(s)$, $\hat{\phi}_Y^{(k)}(t)$, and $\hat{\phi}_{XY}^{(k)}(s, t)$ denote the local empirical characteristic functions computed at client k . Then the global empirical discrepancy $\hat{\Delta}_{XY}(s, t)$ satisfies:*

$$\hat{\Delta}_{XY}(s, t) = \sum_{k=1}^K w_k \hat{\phi}_{XY}^{(k)}(s, t) - \left(\sum_{k=1}^K w_k \hat{\phi}_X^{(k)}(s) \right) \cdot \left(\sum_{k=1}^K w_k \hat{\phi}_Y^{(k)}(t) \right), \quad (15)$$

and the global test statistic $\hat{T}_N = N \cdot \hat{\mathcal{T}}_L(X, Y)$ can be computed **exactly** from the 6KL real-valued local statistics

$$\left\{ \text{Re}[\hat{\phi}_X^{(k)}(s_\ell)], \text{Im}[\hat{\phi}_X^{(k)}(s_\ell)], \text{Re}[\hat{\phi}_Y^{(k)}(t_\ell)], \text{Im}[\hat{\phi}_Y^{(k)}(t_\ell)], \right. \\ \left. \text{Re}[\hat{\phi}_{XY}^{(k)}(s_\ell, t_\ell)], \text{Im}[\hat{\phi}_{XY}^{(k)}(s_\ell, t_\ell)] \right\}_{k \in [K], \ell \in [L]} \quad (16)$$

together with the sample sizes $\{n_k\}_{k=1}^K$, without any access to raw data. A detailed proof is in [Appendix D.1](#)¹.

Test Statistic and Asymptotic Distribution. We define the *federated CF independence test statistic* as:

$$\hat{T}_N \triangleq N \cdot \frac{1}{L} \sum_{\ell=1}^L |\hat{\Delta}_{XY}(s_\ell, t_\ell)|^2 \quad (17)$$

$$= \frac{N}{L} \sum_{\ell=1}^L \left[(\text{Re}[\hat{\Delta}_{XY}(s_\ell, t_\ell)])^2 + (\text{Im}[\hat{\Delta}_{XY}(s_\ell, t_\ell)])^2 \right]. \quad (18)$$

Expanding the squared modulus, we obtain:

$$|\hat{\Delta}_{XY}(s_\ell, t_\ell)|^2 = (\hat{C}_{XY}^\ell - \hat{C}_X^\ell \hat{C}_Y^\ell + \hat{S}_X^\ell \hat{S}_Y^\ell)^2 + (\hat{S}_{XY}^\ell - \hat{S}_X^\ell \hat{C}_Y^\ell - \hat{C}_X^\ell \hat{S}_Y^\ell)^2, \quad (19)$$

where we use the shorthand $\hat{C}_X^\ell = \text{Re}[\hat{\phi}_X(s_\ell)]$, $\hat{S}_X^\ell = \text{Im}[\hat{\phi}_X(s_\ell)]$, and analogously for Y and the joint.

THEOREM 3.3. *Under the null hypothesis $H_0 : X \perp\!\!\!\perp Y$ and standard regularity conditions (finite fourth moments), as $N \rightarrow \infty$ with L fixed:*

$$\hat{T}_N \xrightarrow{d} \sum_{\ell=1}^L \lambda_\ell \chi_1^2(\ell), \quad (20)$$

where $\{\lambda_\ell\}_{\ell=1}^L$ are the eigenvalues of the covariance matrix of the $2L$ -dimensional vector $(\text{Re}[\hat{\Delta}_{XY}(s_\ell, t_\ell)], \text{Im}[\hat{\Delta}_{XY}(s_\ell, t_\ell)])_{\ell=1}^L$ and $\chi_1^2(\ell)$ are independent chi-squared random variables with one degree of freedom. Under the simplifying assumption that the eigenvalues are approximately equal (i.e., $\lambda_\ell \approx \bar{\lambda}$ for all ℓ), the distribution is well-approximated by a scaled chi-squared: $\hat{T}_N \sim \text{Gamma}(\frac{L}{2}, \frac{2}{N})$, where $\text{Gamma}(k, \theta)$ denotes the Gamma distribution with shape k and scale θ . A detailed proof is in [Appendix D.2](#)

Alternatively, when the asymptotic approximation is insufficiently accurate (e.g., for small N or heavy-tailed data), we employ a *federated permutation test* to obtain exact finite-sample p -values.

Federated Permutation Test. The permutation test generates the null distribution of $\hat{T}_N$ by breaking the dependence between X and Y through random permutation of Y -indices. In the federated setting, each client k independently permutes its local Y -samples using a shared random seed π_b for permutation $b \in [B]$, computes the permuted joint CF statistics $\hat{\phi}_{XY\pi_b}^{(k)}(s_\ell, t_\ell)$, and transmits them to the server. The server aggregates to obtain the permuted global statistic $\hat{T}_N^{(b)}$ for each permutation b , and the p -value is:

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbb{1}[\hat{T}_N^{(b)} \geq \hat{T}_N]}{1 + B}. \quad (21)$$

Remark 1 (Validity of Local Permutation). *Under Assumption 4, the null hypothesis $H_0 : X \perp\!\!\!\perp Y \mid Z$ implies that X and Y are conditionally independent within each client. Therefore, permuting Y -indices locally at each client preserves the null distribution. The aggregated permuted statistic $\hat{T}_N^{(b)}$ is a valid draw from the null distribution of $\hat{T}_N$, ensuring the type-I error control of the permutation test. A formal justification is provided in [Appendix D.9](#).*

3.1.2 Extension to Conditional Independence Testing. To test $X \perp\!\!\!\perp Y \mid Z$ for a conditioning set $Z = \{Z_1, \dots, Z_m\}$, we adopt the *nonlinear residualization strategy* (Definition 2.3). Each client k locally computes residuals:

$$R_X^{(k)} = X^{(k)} - \hat{g}_X^{(k)}(Z^{(k)}), \quad R_Y^{(k)} = Y^{(k)} - \hat{g}_Y^{(k)}(Z^{(k)}), \quad (22)$$

where $\hat{g}_X^{(k)}$ and $\hat{g}_Y^{(k)}$ are nonparametric regression estimators (we use k -nearest neighbors regression) fitted on client k 's local data. The independence test then reduces to testing marginal independence $R_X \perp\!\!\!\perp R_Y$ using the CF discrepancy test on the residuals.

Lemma 3.4. *Under Assumptions 2–5, if $\hat{g}_X^{(k)}$ and $\hat{g}_Y^{(k)}$ are universally consistent (i.e., $\mathbb{E}[\|\hat{g}_X^{(k)}(Z) - g_X(Z)\|^2] \rightarrow 0$ as $n_k \rightarrow \infty$ for all k), then the federated CF discrepancy test applied to the aggregated residual statistics is consistent: as $\min_k n_k \rightarrow \infty$,*

$$\mathbb{P}(\text{reject } H_0 \mid H_0 \text{ true}) \leq \alpha + o(1), \quad \mathbb{P}(\text{reject } H_0 \mid H_1 \text{ true}) \rightarrow 1, \quad (23)$$

where α is the significance level. A detailed proof is in [Appendix D.3](#).

3.1.3 Incremental PC Set Discovery. For each target variable V_i , we learn its PC set through a two-stage incremental procedure inspired by the HITON-PC algorithm [1], adapted to the federated characteristic function testing framework.

Stage I: Marginal Association Screening. For every candidate variable $V_j \in \mathcal{V} \setminus \{V_i\}$, the server requests each client to compute the marginal CF statistics for the pair (V_j, V_i) . The server aggregates these statistics via Eq. (14) and evaluates the marginal independence test $V_j \perp\!\!\!\perp V_i$ using the test statistic in Eq. (17). Variables that are *not* marginally independent of V_i (i.e., p -value $\leq \alpha$) are admitted into the *candidate PC set* C_i , ordered by decreasing association strength:

$$C_i = \left\{ V_j \in \mathcal{V} \setminus \{V_i\} : \hat{p}_{j,i}^{(0)} \leq \alpha \right\}, \quad \text{ordered by } \hat{T}_N(V_j, V_i) \downarrow. \quad (24)$$

Stage II: Conditional Pruning with Incremental Conditioning. Starting from C_i , we iteratively test whether each variable $V_j \in C_i$ can be separated from V_i by conditioning on subsets of the current candidate set. Variables are processed in decreasing order of marginal association strength, and for each newly added variable V_j , we perform backward elimination: For each V_a in the current candidate

¹Appendix D is available at <https://xianjie-guo.github.io/pdf/DC-FNCD-app.pdf>.

set $\text{PC}_i^{\text{cand}}$ (in reverse order of addition), we search for a separating set $\mathcal{S} \subseteq \text{PC}_i^{\text{cand}} \setminus \{V_a\}$ such that $V_a \perp\!\!\!\perp V_i \mid \mathcal{S}$. The conditioning set size is incrementally increased from 0 to $\min(|\text{PC}_i^{\text{cand}}| - 1, d_{\max})$, where $d_{\max}$ is a user-specified maximum conditioning depth.

Proposition 3.5. *Let $\hat{\text{PC}}(V_i)$ and $\hat{\text{PC}}(V_j)$ denote the estimated PC set for variables V_i and V_j , where $j \neq i$. In the divide-and-conquer architecture of DC-FNCD, the estimation errors satisfy:*

$$\mathbb{P}(\hat{\text{PC}}(V_i) \neq \text{PC}_{\mathcal{G}^*}(V_i) \mid \hat{\text{PC}}(V_j) \neq \text{PC}_{\mathcal{G}^*}(V_j)) = \mathbb{P}(\hat{\text{PC}}(V_i) \neq \text{PC}_{\mathcal{G}^*}(V_i)), \quad (25)$$

i.e., estimation errors are statistically independent across different target variables. This contrasts with monolithic global methods where errors propagate through the global conditioning structure.

A detailed proof is provided in [Appendix D.4](#).

3.2 Phase 2: Conflict-Aware Skeleton Construction

After Phase 1, the server obtains p PC sets $\{\hat{\text{PC}}(V_i)\}_{i=1}^p$. Since each PC set is learned independently, inconsistencies may arise: $V_j \in \hat{\text{PC}}(V_i)$ does not necessarily imply $V_i \in \hat{\text{PC}}(V_j)$. We construct the global skeleton through a principled conflict resolution mechanism.

Definition 3.6 (Symmetric and Asymmetric Edges). *An edge (V_i, V_j) with $i < j$ is called symmetric if $V_j \in \hat{\text{PC}}(V_i) \wedge V_i \in \hat{\text{PC}}(V_j)$, and asymmetric if exactly one inclusion holds:*

$$\mathcal{E}_{\text{sym}} \triangleq \{(V_i, V_j) : i < j, V_j \in \hat{\text{PC}}(V_i) \wedge V_i \in \hat{\text{PC}}(V_j)\}, \quad (26)$$

$$\mathcal{E}_{\text{asym}} \triangleq \{(V_i, V_j) : i < j, V_j \in \hat{\text{PC}}(V_i) \oplus V_i \in \hat{\text{PC}}(V_j)\}, \quad (27)$$

where $\oplus$ denotes the exclusive-or operation.

Symmetric edges reflect consistent evidence from both local neighborhoods and are directly included in the skeleton. Asymmetric edges represent conflicts that require resolution.

ANM-Based Conflict Resolution. For each asymmetric edge $(V_i, V_j) \in \mathcal{E}_{\text{asym}}$, the server computes two federated ANM test statistics (detailed in Section 3.3) measuring the strength of residual dependence under both causal directions:

$$\hat{T}_{i,j}^{\rightarrow} = \hat{T}_N(R_{V_j|V_i}, V_i), \quad \hat{T}_{i,j}^{\leftarrow} = \hat{T}_N(R_{V_i|V_j}, V_j), \quad (28)$$

where $R_{V_j|V_i} = V_j - \hat{f}(V_i)$ denotes the regression residual. An asymmetric edge is *confirmed* if the evidence of association significantly exceeds random fluctuation:

$$(V_i, V_j) \in \hat{\mathcal{E}}_{\text{skeleton}} \iff \frac{\max(\hat{T}_{i,j}^{\rightarrow}, \hat{T}_{i,j}^{\leftarrow})}{\min(\hat{T}_{i,j}^{\rightarrow}, \hat{T}_{i,j}^{\leftarrow}) + \epsilon} > \tau, \quad (29)$$

where $\tau > 1$ is a confirmation threshold (default $\tau = 2$) and $\epsilon > 0$ is a small constant for numerical stability. The intuition is that a true edge should exhibit substantial residual dependence in at least one direction, while a spurious edge (arising from, e.g., limited local samples in one neighborhood) should show weak evidence in both directions. The final skeleton is:

$$\hat{\mathcal{E}}_{\text{skeleton}} = \mathcal{E}_{\text{sym}} \cup \{(V_i, V_j) \in \mathcal{E}_{\text{asym}} : \text{confirmed by Eq. (29)}\}. \quad (30)$$

THEOREM 3.7. *Under Assumptions 1–5, if the significance level $\alpha = \alpha_N \rightarrow 0$ and $\alpha_N \cdot N / \log N \rightarrow \infty$ as $\min_k n_k \rightarrow \infty$, then the skeleton $\hat{\mathcal{E}}_{\text{skeleton}}$ recovered by DC-FNCD is consistent:*

$$\mathbb{P}(\hat{\mathcal{E}}_{\text{skeleton}} = \mathcal{E}_{\text{skeleton}}^*) \rightarrow 1 \quad \text{as } \min_k n_k \rightarrow \infty, \quad (31)$$

where $\mathcal{E}_{\text{skeleton}}^*$ denotes the true skeleton of $\mathcal{G}^*$. Proof in [Appendix D.5](#).

3.3 Phase 3: Federated ANM-Based Orientation

Given the learned skeleton, we determine edge directions using the Additive Noise Model (ANM) principle [15]: under Assumption 3, if $V_i \rightarrow V_j$ in the true DAG, then $V_j = f(V_i) + \varepsilon_j$ with $\varepsilon_j \perp\!\!\!\perp V_i$, while in general $V_i \neq g(V_j) + \eta_i$ with $\eta_i \perp\!\!\!\perp V_j$ for the reverse direction.

Step 1: Local Residual Computation. Each client k locally computes the regression residuals for both causal directions:

$$R_{j|i}^{(k)} = V_j^{(k)} - \hat{f}_{j|i}^{(k)}(V_i^{(k)}), \quad (\text{forward: } V_i \rightarrow V_j) \quad (32)$$

$$R_{i|j}^{(k)} = V_i^{(k)} - \hat{f}_{i|j}^{(k)}(V_j^{(k)}), \quad (\text{backward: } V_j \rightarrow V_i) \quad (33)$$

where $\hat{f}_{j|i}^{(k)}$ and $\hat{f}_{i|j}^{(k)}$ are k -nearest neighbor regressors.

Step 2: Local CF Statistics of Residuals. Client k computes the characteristic function statistics for the residual-cause pair: $(\hat{\phi}_{R_{j|i}, V_i}^{(k)}, \hat{\phi}_{V_i}^{(k)})$ for the forward direction, and analogously for backward.

Step 3: Server Aggregation and Direction Decision. The server aggregates the CF statistics and computes the test statistics:

$$\hat{T}_{i,j}^{\rightarrow} = N \cdot \frac{1}{L} \sum_{\ell=1}^L \left| \hat{\phi}_{R_{j|i}, V_i}(s_{\ell}, t_{\ell}) - \hat{\phi}_{R_{j|i}}(s_{\ell}) \cdot \hat{\phi}_{V_i}(t_{\ell}) \right|^2, \quad (34)$$

$$\hat{T}_{i,j}^{\leftarrow} = N \cdot \frac{1}{L} \sum_{\ell=1}^L \left| \hat{\phi}_{R_{i|j}, V_j}(s_{\ell}, t_{\ell}) - \hat{\phi}_{R_{i|j}}(s_{\ell}) \cdot \hat{\phi}_{V_j}(t_{\ell}) \right|^2. \quad (35)$$

The direction is assigned by comparing the residual scores:

$$\text{Orient} \quad \begin{cases} V_i \rightarrow V_j & \text{if } \hat{T}_{i,j}^{\rightarrow} \leq \hat{T}_{i,j}^{\leftarrow}, \\ V_j \rightarrow V_i & \text{if } \hat{T}_{i,j}^{\rightarrow} > \hat{T}_{i,j}^{\leftarrow}. \end{cases} \quad (36)$$

The rationale is that the *correct* causal direction yields residuals that are more independent of the cause variable, while the *reverse* direction produces residuals that retain spurious dependence.

THEOREM 3.8. *Under Assumptions 3 and 4, if the true causal model $V_j = f(V_i) + \varepsilon_j$ satisfies the ANM identifiability conditions of Hoyer et al. [15], and the local KNN regressors are universally consistent, then for each edge (V_i, V_j) with true direction $V_i \rightarrow V_j$:*

$$\mathbb{P}(\hat{T}_{i,j}^{\rightarrow} < \hat{T}_{i,j}^{\leftarrow}) \rightarrow 1 \quad \text{as } \min_k n_k \rightarrow \infty. \quad (37)$$

A detailed proof is provided in [Appendix D.6](#). Combining Theorems 3.7 and 3.8, we obtain the overall consistency of DC-FNCD:

Corollary 3.9 (Global Consistency of DC-FNCD). *Under Assumptions 1–5 and the ANM identifiability conditions, DC-FNCD recovers the true DAG $\mathcal{G}^*$ with probability tending to 1 as $\min_k n_k \rightarrow \infty$.*

3.4 Privacy and Communication Analysis

Privacy Guarantees. We now formally analyze the privacy properties of DC-FNCD. The fundamental privacy mechanism is the *characteristic function barrier*: clients transmit only empirical CF statistics—averaged trigonometric transforms of their data—rather than raw samples or model parameters that could be inverted.

THEOREM 3.10. *Let $\mathcal{S}^{(k)} = \{(\text{Re}[\hat{\phi}_{(\cdot)}^{(k)}(s_t)], \text{Im}[\hat{\phi}_{(\cdot)}^{(k)}(t_t)])\}_{t=1}^L$ denote the complete set of CF statistics transmitted by client k across all phases of DC-FNCD. Then: (1) The mapping from local data $\mathbf{D}^{(k)} \in \mathbb{R}^{n_k \times p}$ to $\mathcal{S}^{(k)} \in \mathbb{R}^{O(Lp^2)}$ is a many-to-one function when $n_k \gg Lp^2$, i.e., infinitely many datasets produce identical CF statistics. Formally, for any $\mathcal{S}^{(k)}$, the preimage set $\{\mathbf{D} : \Phi(\mathbf{D}) = \mathcal{S}^{(k)}\}$ is uncountably infinite. (2) Each CF statistic $\hat{\phi}_X^{(k)}(s) = \frac{1}{n_k} \sum_{j=1}^{n_k} e^{isx_j^{(k)}}$ is a sample mean of bounded random variables ($|e^{isx}| = 1$), and by standard results in statistical estimation, the mutual information between $\mathbf{D}^{(k)}$ and $\mathcal{S}^{(k)}$ satisfies: $I(\mathbf{D}^{(k)}; \mathcal{S}^{(k)}) \leq O\left(\frac{Lp^2}{n_k}\right)$. (3) Since the server only requires the weighted sum $\sum_k w_k \hat{\phi}^{(k)}(\cdot)$, standard secure aggregation protocols [3] can be applied to prevent the server from observing individual client statistics, further strengthening privacy.*

A detailed proof is provided in Appendix D.7.

Communication and Time Complexity. We analyze the communication cost of DC-FNCD in terms of the number of real-valued scalars transmitted between clients and the server.

Proposition 3.11. *The total communication cost of DC-FNCD across all K clients is:*

$$C_{\text{comm}} = O(K \cdot L \cdot N_{\text{CI}} \cdot (1 + B)), \quad (38)$$

where L is the number of frequency points, N_{CI} is the total number of CI tests performed, and B is the number of permutations (set to 0 if asymptotic p -values are used). In each CI test, each client transmits $6L$ real values (3 complex-valued CF statistics at L frequencies), plus optionally $2LB$ values for permutation statistics. For the divide-and-conquer PC learning with maximum conditioning depth d_{max} , the number of CI tests satisfies $N_{\text{CI}} = O(p \cdot \binom{|C_{\text{max}}|}{d_{\text{max}}})$, where $|C_{\text{max}}|$ is the maximum candidate PC set size. In sparse graphs with bounded degree d , $|C_{\text{max}}| = O(d)$, yielding:

$$C_{\text{comm}} = O(K \cdot L \cdot p \cdot d^{d_{\text{max}}} \cdot (1 + B)), \quad (39)$$

which scales linearly in p , compared to the $O(p^2)$ or $O(p^3)$ scaling of monolithic global methods that iterate over all variable pairs with growing conditioning sets. A detailed proof is in Appendix D.8.

Proposition 3.12. *The overall time complexity of DC-FNCD is:*

$$\begin{aligned} \tau_{\text{total}} = & \underbrace{O\left(p \cdot N_{\text{CI}}^{(i)} \cdot (Kn_{\text{max}}L + Kn_{\text{max}} \log n_{\text{max}})\right)}_{\text{Phase 1: Local PC Learning}} \\ & + \underbrace{O(|\mathcal{E}_{\text{asym}}| \cdot Kn_{\text{max}}L)}_{\text{Phase 2: Skeleton}} + \underbrace{O(|\hat{\mathcal{E}}_{\text{skeleton}}| \cdot Kn_{\text{max}}L)}_{\text{Phase 3: Orientation}}, \end{aligned} \quad (40)$$

where $n_{\text{max}} = \max_k n_k$, $N_{\text{CI}}^{(i)}$ is the average number of CI tests per target variable, and the $n_{\text{max}} \log n_{\text{max}}$ term arises from k -NN regression for residual computation. Since the p local PC learning tasks

are embarrassingly parallel, the wall-clock time of Phase 1 can be reduced by a factor of p with parallel execution.

4 Experiments

4.1 Experimental Settings

Datasets. We evaluate DC-FNCD on both synthetic and real-world datasets. For synthetic data, we consider two federated scenarios: (i) **Homogeneous Nonlinear (IID)**: all K clients generate data from the same nonlinear structural equation model (SEM) with Gaussian Process (GP) mechanisms, where each client holds 500–600 i.i.d. samples drawn from a shared causal model; and (ii) **Heterogeneous Nonlinear (Non-IID)**: different clients employ heterogeneous nonlinear mechanisms—randomly assigned from {MLP, GP, MIM, GP-additive}—with varying noise scales ($\sigma \in \{0.8, 1.0\}$), where each client holds 500–600 samples. In both settings, the underlying DAG is an Erdős-Rényi (ER) graph with expected degree 4 (i.e., $|\mathcal{E}| = 2p$) and edge weights sampled uniformly from $[\pm 0.5, \pm 2.0]$. We vary the number of variables $d \in \{10, 15, 20, 25, 40\}$ with $K=10$ clients fixed, and the number of clients $K \in \{3, 5, 10, 18, 30\}$ with $d=20$ fixed. Each configuration is repeated over 5 random seeds. For **real-world evaluation**, we use the **fMRI Hippocampus** dataset [24], which contains BOLD time-series recordings from 6 hippocampal subregions. Following Li et al. [19], we construct a ground-truth DAG with 7 edges based on established neuroanatomical connectivity. We partition the multi-session recordings into $K \in \{3, 5, 10, 18\}$ clients and report results over 3 independent trials.

Comparison Baselines. We compare against five representative FCD methods: **FedPC** [22], which federates the PC algorithm using Fisher- z tests; **FedPC-KBT**, a variant of FedPC using kernel-based independence testing with random Fourier features; **FedCDH** [19], which employs federated constraint-based discovery with heterogeneity handling; **FedACD** [13], an adaptive federated method combining constraint- and score-based strategies; and **NOTEARS-ADMM** [22], which adapts the continuous optimization framework of NOTEARS [41] to the federated setting via ADMM and MLP.

Evaluation Metrics. We employ four DAG-level metrics: **F1 score**, **SHD**, **TPR**, and **FDR** [8, 11]. See Appendix E.1² for the details.

Table 2: Hyperparameters of DC-FNCD and their default values.

Symbol	Description	Default
α	Significance level for CI tests	0.05
L	Number of random frequency points	50–100
σ_ω	Frequency bandwidth	1.0
d_{max}	Maximum conditioning set size	3
B	Number of permutations	200
τ	Confirmation threshold for asymmetric edges	2.0
k_{nn}	Number of neighbors in KNN regression	5
ϵ	Numerical stability constant	10^{-8}

Implementation Details. All experiments are conducted on a workstation with an Intel Core Ultra 7 265K processor, an NVIDIA RTX 5070 Ti GPU (16GB VRAM), and 48GB of system memory.

²Appendix E is available at <https://xianjie-guo.github.io/pdf/DC-FNCD-app.pdf>.

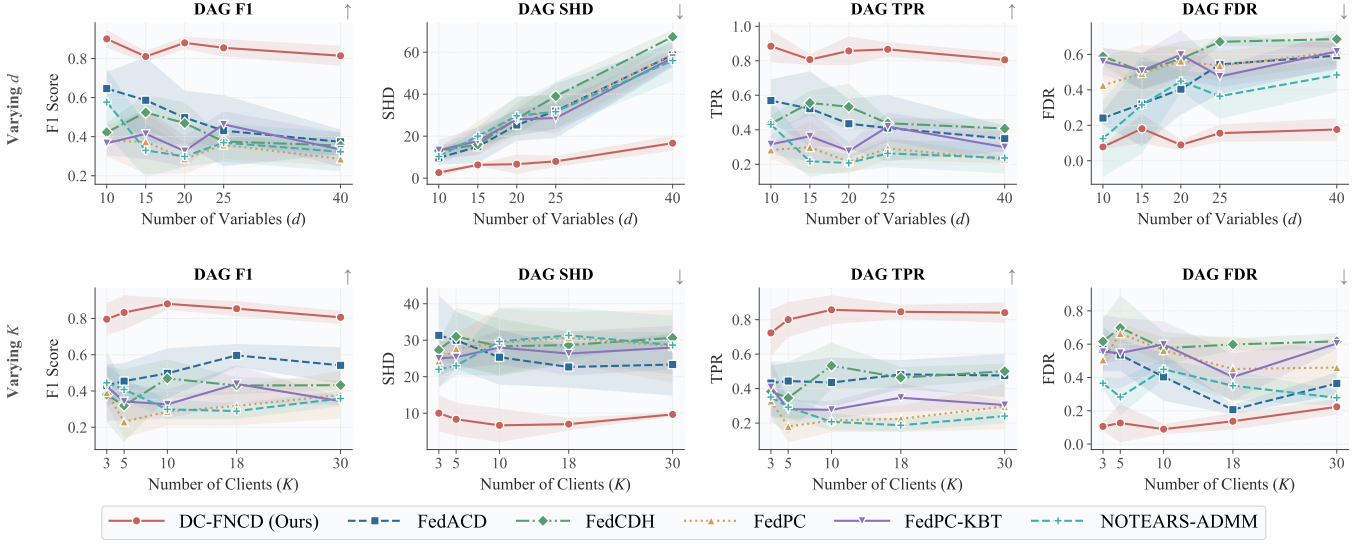


Figure 3: DAG-level metrics on homogeneous nonlinear synthetic data with ER-GP graphs. Top row: varying $d \in \{10, 15, 20, 25, 40\}$ with $K=10$. Bottom row: varying $K \in \{3, 5, 10, 18, 30\}$ with $d=20$. Shaded regions denote ± 1 std over 5 seeds.

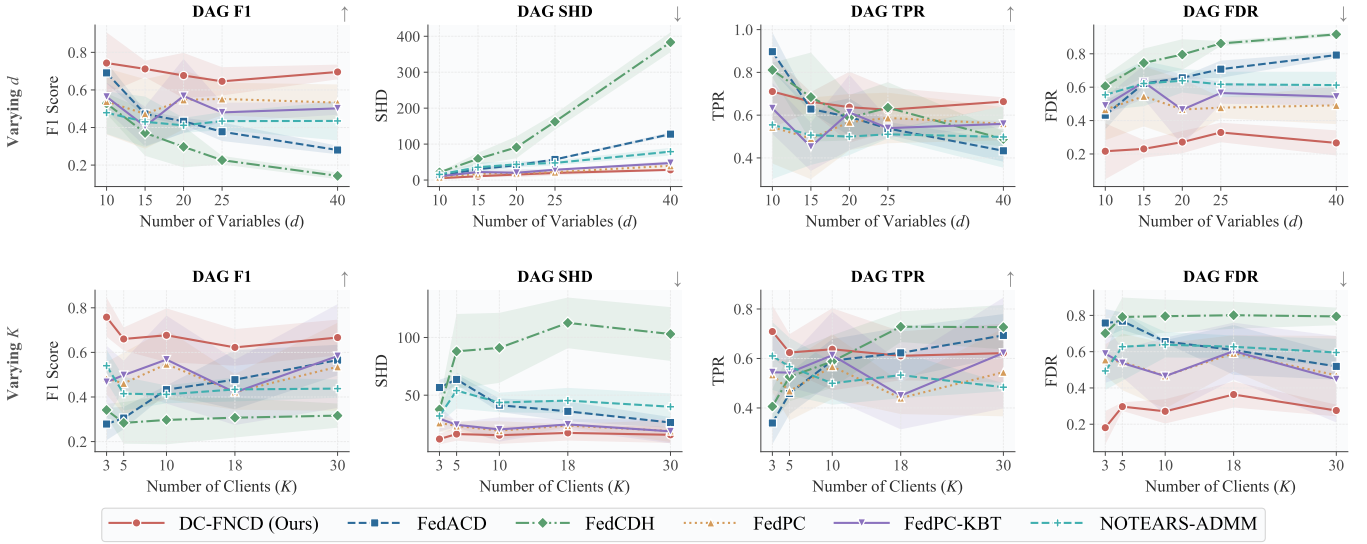


Figure 4: DAG-level metrics on heterogeneous nonlinear (Non-IID) synthetic data with ER graphs. Top row: varying $d \in \{10, 15, 20, 25, 40\}$ with $K=10$. Bottom row: varying $K \in \{3, 5, 10, 18, 30\}$ with $d=20$. Shaded regions denote ± 1 std over 5 seeds.

Random seeds are fixed throughout for reproducibility. Table 2 summarizes the hyperparameters of DC-FNCD and their default values used in our experiments.

4.2 Results on Homogeneous Nonlinear Case

Scalability with graph size. Figure 3 presents the DAG-level metrics under the homogeneous nonlinear setting. We can see that as d increases (top row), DC-FNCD consistently achieves the highest F1 and lowest FDR. At $d=40$, DC-FNCD maintains F1 above 0.8 while the SOTA baseline FedACD drops below 0.40. This advantage stems from the error compartmentalization property (Proposition 3.5):

independent local PC learning prevents error propagation across subproblems, unlike global methods where early CI testing errors cascade through subsequent conditioning.

Scalability with number of clients. As K grows (bottom row), DC-FNCD shows a steady upward trend in F1 and TPR, effectively leveraging additional distributed data through the *lossless* CF aggregation (Lemma 3.2). In contrast, NOTEARS-ADMM degrades at larger K due to consensus optimization difficulties, and FedPC-KBT suffers from accumulated random feature approximation errors.

Linear vs. nonlinear testing. FedPC (Fisher-z tests) deteriorates sharply as d grows, since linear correlation cannot capture

GP-generated nonlinear dependencies. FedPC-KBT mitigates this via kernel-based testing but introduces approximation errors that compound across clients. DC-FNCD avoids both limitations through exact nonparametric CF-based testing.

4.3 Results on Heterogeneous Nonlinear Case

Figure 4 reports results under the more challenging Non-IID setting, where clients employ different nonlinear mechanisms and noise scales. We can see that DC-FNCD achieves the highest F1 and lowest FDR across nearly all configurations. This robustness stems from two design choices: (i) the CF discrepancy test (Definition 3.1) is distribution-free, making it inherently robust to heterogeneous mechanisms; and (ii) local KNN residualization (Section 3.1.2) adapts to each client’s data distribution before aggregation.

Impact of client diversity. Increasing K provides a dual benefit: larger total sample size and mechanism diversity that implicitly regularizes ANM-based orientation. Since ANM identifiability holds generically [15], aggregating residual scores across diverse clients strengthens the orientation signal.

4.4 Results on Real-World Data

Table 3 presents the per-client-partition results on the fMRI Hippocampus dataset [24].

DC-FNCD achieves the best F1 across all client partitions. Across all four settings ($K \in \{3, 5, 10, 18\}$), DC-FNCD attains the highest F1 score, ranging from 0.357 ($K=3$) to 0.478 ($K=10$), and consistently maintains the lowest FDR. The improvement over the second-best method is substantial: at $K=10$, DC-FNCD outperforms FedACD by 7.7% in F1 and reduces FDR by 6.4 percentage points.

Performance improves with more clients. A notable trend is that DC-FNCD’s F1 increases monotonically from $K=3$ to $K=10$, confirming that the lossless CF aggregation mechanism effectively harnesses distributed data. The slight decrease at $K=18$ ($F1 = 0.445$) is attributable to the very small per-client sample size when the fixed dataset is split across 18 clients, which degrades local KNN regression quality.

Baselines struggle on real data. NOTEARS-ADMM exhibits catastrophic failure at $K=3$ ($F1 = 0.199$), as the ADMM consensus optimization fails to converge with extremely small per-client samples. FedPC-KBT consistently underperforms FedPC, suggesting that random Fourier feature approximation introduces more harm than benefit in this low-sample regime. FedACD provides competitive TPR but at the cost of substantially higher FDR, indicating excessive false discoveries. The overall global summary and additional analysis are provided in [Appendix E.4](#)³.

5 Conclusions

We presented DC-FNCD, a divide-and-conquer framework for federated nonlinear causal discovery that addresses the catastrophic error propagation of monolithic global learning approaches. By decomposing structure learning into independent per-variable neighborhood tasks, DC-FNCD isolates estimation errors rather than propagating them globally—critical when clients hold limited, heterogeneous data. Our characteristic function-based independence test

Table 3: fMRI Hippocampus — Per-Client Comparison.

Algorithm	F1 ($\uparrow$)	TPR ($\uparrow$)	FDR ($\downarrow$)
<i>K = 3 Clients</i>			
DC-FNCD (Ours)	0.357±0.06	<u>0.333±0.08</u>	0.613±0.05
FedACD	<u>0.291±0.10</u>	0.381±0.16	<u>0.763±0.06</u>
FedCDH	0.219±0.08	0.238±0.08	0.796±0.08
FedPC	0.251±0.07	<u>0.333±0.08</u>	0.798±0.06
FedPC-KBT	0.127±0.06	0.191±0.08	0.904±0.05
NOTEARS-ADMM	0.199±0.06	0.238±0.08	0.830±0.05
<i>K = 5 Clients</i>			
DC-FNCD (Ours)	0.429±0.14	0.429±0.14	0.571±0.14
FedACD	0.328±0.08	0.429±0.14	0.732±0.04
FedCDH	0.274±0.20	0.286±0.14	0.714±0.27
FedPC	0.203±0.19	0.238±0.22	0.822±0.17
FedPC-KBT	0.207±0.01	0.286±0.00	0.838±0.01
NOTEARS-ADMM	<u>0.345±0.05</u>	0.381±0.08	<u>0.683±0.03</u>
<i>K = 10 Clients</i>			
DC-FNCD (Ours)	0.478±0.10	0.571±0.14	0.587±0.08
FedACD	<u>0.401±0.07</u>	0.476±0.08	0.651±0.07
FedCDH	0.386±0.12	<u>0.524±0.16</u>	0.694±0.10
FedPC	0.363±0.23	0.476±0.30	0.707±0.19
FedPC-KBT	0.134±0.06	0.191±0.08	0.897±0.04
NOTEARS-ADMM	0.387±0.10	0.429±0.14	<u>0.641±0.08</u>
<i>K = 18 Clients</i>			
DC-FNCD (Ours)	0.445±0.02	0.571±0.00	0.634±0.03
FedACD	0.360±0.01	0.429±0.00	0.689±0.02
FedCDH	0.349±0.05	<u>0.524±0.08</u>	0.737±0.04
FedPC	<u>0.409±0.19</u>	<u>0.524±0.22</u>	<u>0.664±0.16</u>
FedPC-KBT	0.137±0.06	0.191±0.08	0.893±0.05
NOTEARS-ADMM	0.225±0.07	0.238±0.08	0.786±0.06

achieves exact decomposition across clients without kernel approximation, enabling lossless aggregation of nonlinear dependencies. The framework provides formal guarantees on consistency, privacy, and $O(p)$ -factor communication reduction, with experiments demonstrating improvements over state-of-the-art baselines under both homogeneous and heterogeneous nonlinear settings.

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³Appendix E is available at <https://xianjie-guo.github.io/pdf/DC-FNCD-app.pdf>.

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APPENDIX

A Extended Related Work

Federated causal discovery (FCD) has rapidly evolved as a research area at the intersection of causal inference and distributed learning. We provide a comprehensive review, organized into two major categories based on the type of causal mechanisms assumed.

A.1 Federated Linear Causal Discovery

Methods in this category assume linear structural equation models (SEMs), i.e., $V_i = \sum_{j \in \text{Pa}(V_i)} w_{ij} V_j + \varepsilon_i$, which enable closed-form sufficient statistics and gradient computation.

Score-Based Methods. NOTEARS-ADMM [22] is an early and influential work that adapted the continuous optimization framework of NOTEARS [41] to the federated setting via the alternating direction method of multipliers (ADMM). The method formulates DAG learning as a constrained optimization problem

$$\min_{\mathbf{W}} \sum_k \mathcal{L}^{(k)}(\mathbf{W}) \text{ s.t. } h(\mathbf{W}) = 0,$$

where $h(\mathbf{W}) = \text{tr}(e^{\mathbf{W} \circ \mathbf{W}}) - p$ is the acyclicity constraint. ADMM decomposes this into local subproblems where each client optimizes a proximal objective involving its local data, and a global consensus step that enforces consistency and acyclicity. Only the model parameters $\mathbf{W}$ (the weighted adjacency matrix) are exchanged, preserving data privacy. While effective for linear models, the ADMM decomposition couples all edges through the acyclicity constraint, making it susceptible to the error propagation issue identified in our work.

DARLS [37] (Distributed Annealing on Regularized Likelihood Score) proposed a fundamentally different search strategy: instead of continuous relaxation, it searches over the space of topological sorts via simulated annealing, where the optimal graph compatible with each sort is found by distributed optimization. A key theoretical contribution of DARLS is the finite-sample oracle guarantee: the federated estimate converges to the centralized oracle solution at rate $O(\log(n)/\sqrt{m})$, where n is the total sample size and m is the smallest local sample size. DARLS also introduced identifiability results for DAGs parameterized by generalized linear models (GLMs), extending beyond the standard linear Gaussian setting. However, the simulated annealing search over permutations incurs substantial computational cost, particularly for large p .

FedCausal [36] designed a unified global optimization formula that naturally aggregates local causal graphs while constraining global acyclicity. The key innovation is a flexible optimization objective that adaptively handles both homogeneous and heterogeneous data distributions. The method formulates the federated DAG learning as $\min_{\mathbf{W}} \sum_k w_k \|\mathbf{D}^{(k)} - \mathbf{D}^{(k)} \mathbf{W}\|_F^2 + \lambda \|\mathbf{W}\|_1$ subject to $h(\mathbf{W}) = 0$, where the weighted aggregation of local least-squares losses is shown to be equivalent to the global loss on pooled data under homogeneous distributions. For heterogeneous settings, FedCausal introduces client-specific scaling factors. While the linear version is well-studied, the nonlinear extension relies on neural network parameterization with less theoretical grounding.

FedGES [33] presented a federated adaptation of the Greedy Equivalence Search (GES) algorithm, where clients run GES locally with an edge-growth limit and the server aggregates structures using thresholded consensus fusion or min-cut-based consensus. A distinctive feature is that FedGES exchanges only DAG structures (not sufficient statistics or model parameters), making it inherently privacy-enhancing. The method preserves the theoretical guarantees of centralized GES with score-consistent metrics and ensures finite termination. Experiments on benchmark networks with up to 724 nodes and 100 clients demonstrate scalability, though the method is primarily designed for discrete Bayesian networks with linear scoring functions.

PERI [20] took a game-theoretic perspective, formulating federated causal discovery as a distributed min-max regret optimization

problem. The max-regret scoring criterion is proven to be consistent and can be used within GES for federated structure learning. A notable privacy advantage is that PERI avoids sharing locally discovered causal structures in every iteration (unlike most alternatives), thereby reducing both communication costs and information leakage. However, the min-max formulation introduces conservatism that may reduce statistical power in detecting weak causal effects.

Constraint-Based Methods. FedC2SL [35] proposed a federated constraint-based scheme with two variants: FedPC for causally sufficient settings and FedFCI for settings with latent confounders. The core innovation is a federated conditional independence test that examines conditional independence without collecting raw data, using Fisher’s z-transformation of partial correlations aggregated across clients. FedC2SL requires weaker assumptions about data distributions than prior methods and offers stronger resistance to data variability among clients. However, the reliance on partial correlations fundamentally limits the method to linear (or monotonic) dependencies.

FedCSL [10] addressed scalability and accuracy limitations of existing federated CSL methods through two strategies: (i) a federated local-to-global learning strategy that decomposes the problem into local subproblems for scalability, and (ii) a weighted aggregation strategy that accounts for varying sample sizes across clients without complex encryption. While the local-to-global idea shares conceptual similarity with our divide-and-conquer approach, FedCSL operates under linear assumptions using partial correlation-based tests and does not address nonlinear causal mechanisms.

Hybrid and Specialized Methods. FedACD [13] (Federated Adaptive Causal Discovery) addressed the problem of *sample quality heterogeneity*, where different clients may have varying data quality across different variable subspaces. During federated aggregation, FedACD adaptively selects causal relationships learned under high-quality variable spaces from each client while masking those from low-quality spaces. This selective aggregation strategy is complementary to our approach and could potentially be integrated into the skeleton merge phase of DC-FNCD.

FedImpCSL [28] tackled the practical challenge of missing data in federated causal structure learning. The method combines a local-to-global missing data imputation strategy with a dynamic client weighting mechanism to address inter-client differences exacerbated by missing values. While addressing an important practical concern, the method assumes linear causal mechanisms.

FedCDnv [34] studied the setting where clients observe overlapping but non-identical variable sets, developing theories for detecting spurious dependencies introduced by non-overlapping variables and a two-level priority selection strategy for aggregating local updates. This addresses an orthogonal challenge to ours and represents an important direction for future extension of DC-FNCD.

A.2 Federated Nonlinear Causal Discovery

Methods in this category handle nonlinear causal mechanisms, which are more realistic but substantially more challenging due to the absence of closed-form sufficient statistics.

Score-Based (Continuous Optimization) Methods. FedDAG [7] is a pioneering gradient-based framework for federated DAG learning

under additive noise models (ANMs). The key architectural innovation is a two-level model structure: the first level learns edges and directions of the graph (shared across clients via server aggregation), while the second level approximates the causal mechanisms (updated locally to accommodate data heterogeneity). The overall learning task is formulated as a continuous optimization problem with the NOTEARS acyclicity constraint, solved by gradient descent. FedDAG elegantly handles heterogeneous data by allowing mechanism parameters to vary across clients while enforcing a shared structure. However, the monolithic optimization over the full adjacency matrix means that gradient updates for one edge are entangled with all others through the acyclicity constraint, creating the error propagation vulnerability that DC-FNCD addresses.

NOTEARS-ADMM (nonlinear version) [22] extended the ADMM-based framework to nonlinear settings by parameterizing causal mechanisms with multi-layer perceptrons (MLPs). Each client maintains a local MLP model, and the ADMM consensus step enforces agreement on the structural parameters (which edges exist) while allowing mechanism parameters to differ. The nonlinear extension inherits the same architectural limitation as the linear version: the global acyclicity constraint couples all structural decisions.

FedCausal (nonlinear version) [36] extended its unified optimization framework to nonlinear settings by replacing the linear reconstruction loss with neural network-parameterized mechanisms. The nonparametric version employs kernel-based regression for mechanism estimation, enabling flexible modeling of nonlinear dependencies. However, the global optimization objective still operates over the entire adjacency matrix simultaneously, and the acyclicity constraint $h(\mathbf{W}) = 0$ continues to entangle all edge decisions. The adaptive weighting mechanism for heterogeneous data provides some robustness, but does not fundamentally address the error propagation issue inherent in the monolithic architecture.

Constraint-Based (Independence Testing) Methods. FedCDH [19] represents the state-of-the-art in constraint-based federated nonlinear causal discovery. It introduces two key components: (i) a federated conditional independence test (FCIT) based on random feature approximation of kernel matrices, where clients compute local random feature covariance matrices that are aggregated at the server; and (ii) a federated independent change principle (FICP) that exploits heterogeneity across clients for causal direction determination by treating the client index as a surrogate variable. The nonparametric nature of both FCIT and FICP makes no assumption about functional forms, enabling the handling of arbitrary causal models. However, the random feature approximation introduces an additional layer of error: the kernel matrix $\mathcal{K} \approx \Phi\Phi^\top$ where Φ is the random feature matrix, and the approximation quality degrades in high-dimensional conditioning sets. Moreover, the skeleton discovery phase follows the standard global PC algorithm, where conditioning sets grow across the entire variable space, and each decision depends on the accumulated history of all prior decisions, making it vulnerable to cascading errors.

FedPC-KBT [16] extended the FedPC framework to nonlinear settings by replacing Fisher’s z -test with a kernel-based conditional independence test (KCI-test) [39]. The layer-wise aggregation strategy of FedPC is preserved: at each conditioning depth ℓ , clients independently execute the KCI-test on local data and transmit test

statistics (or p -values) to the server for aggregation. While conceptually straightforward, the KCI-test has cubic computational complexity $O(n^3)$ in sample size due to kernel matrix eigendecomposition, making it prohibitively expensive for large-scale federated settings. Furthermore, the aggregation of local p -values (rather than sufficient statistics) is statistically suboptimal and can lead to loss of power, particularly when local sample sizes are small and individual tests are underpowered.

B Synthetic Data Generation Details

Graph generation. All synthetic DAGs are generated as Erdős–Rényi (ER) random graphs. For a target graph with d nodes and $|\mathcal{E}|=2d$ edges, we set the edge creation probability to $p_{\text{edge}} = 2|\mathcal{E}|/d^2$, generate an undirected ER graph via

```
networkx.erdos_renyi_graph,
```

and obtain a DAG through random acyclic orientation (applying a random permutation and retaining only lower-triangular entries). Edge weights are sampled uniformly from $[\pm 0.5, \pm 2.0]$, with signs randomized independently.

Nonlinear mechanisms. For the homogeneous (IID) setting, we use Gaussian Process (GP) mechanisms: given parent values $\mathbf{x}_{\text{pa}}$ for a node j , the child value is generated as $x_j = f_{\text{GP}}(\mathbf{x}_{\text{pa}}) + \varepsilon_j$, where f_{GP} is sampled from a GP prior with RBF kernel (length_scale=1.0, output_scale=1.0) and $\varepsilon_j \sim \mathcal{N}(0, 1)$. For computational efficiency, we directly compute the kernel matrix and sample via Cholesky decomposition rather than using iterative GP implementations.

Heterogeneous mechanisms. For the Non-IID setting, each client is randomly assigned one of four nonlinear mechanism types:

- **MLP:** $x_j = \sigma(\mathbf{x}_{\text{pa}}\mathbf{W}_1)\mathbf{w}_2 + \varepsilon_j$, with $\mathbf{W}_1 \in \mathbb{R}^{|\text{pa}| \times 100}$, $\mathbf{w}_2 \in \mathbb{R}^{100}$, weights from $\mathcal{U}[\pm 0.5, \pm 2.0]$, and $\sigma(\cdot)$ the sigmoid function;
- **GP:** Gaussian Process as described above;
- **MIM (Mixed Independent Mechanisms):** $x_j = \tanh(\mathbf{x}_{\text{pa}}\mathbf{w}_1) + \cos(\mathbf{x}_{\text{pa}}\mathbf{w}_2) + \sin(\mathbf{x}_{\text{pa}}\mathbf{w}_3) + \varepsilon_j$;
- **GP-additive:** $x_j = \sum_{i \in \text{pa}(j)} f_{\text{GP}}^{(i)}(x_i) + \varepsilon_j$, with independent GP samples per parent dimension.

Noise scales are randomly assigned from $\{0.8, 1.0\}$ per client. Sample sizes per client are drawn uniformly from $[500, 600]$ for Non-IID and $[500, 600]$ for IID settings.

Experimental configurations. For each combination of (d, K, seed) , we generate independent datasets. The five random seeds used are $\{108, 9965, 555, 16987, 3263\}$. In the “varying d ” experiments, we fix $K=10$ and sweep $d \in \{10, 15, 20, 25, 40\}$. In the “varying K ” experiments, we fix $d=20$ and sweep $K \in \{3, 5, 10, 18, 30\}$. All results in the main text and appendix report the mean $\pm$ standard deviation across these 5 seeds.

C Software and Reproducibility

Our implementation of DC-FNCD is written in Python 3.12 using the following key libraries:

- NumPy (v2.3.5) for array operations and CF computation.
- scikit-learn (v1.2+) for KNN regression (ball_tree algorithm with KNeighborsRegressor).
- SciPy (v1.16.3) for Gamma distribution CDF evaluation.
- joblib for parallel execution of client computations.